

ANSWER KEY

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	4	2	3	4	4	4	1	2	1	3	1	3	2	4	2	4	2	1	4	3
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	4	3	2	2	2	2	4	4	1	1	4	4	2	4	4	1	3	3	1	1
Q.	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A.	4	1	2	4	3	1	2	4	2	3	2	3	3	1	4	2	1	3	1	1
Q.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80
A.	1	2	2	4	3	3	1	1	2	3	3	3	2	1	4	4	2	3	1	3
Q.	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100
A.	3	4	3	4	4	2	1	1	3	3	3	4	1	2	2	2	2	3	2	1
Q.	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
A.	3	2	3	1	3	3	1	2	1	3	2	1	1	3	4	3	2	4	4	1
Q.	121	122	123	124	125	126	127	128	129	130										
A.	1	4	2	4	4	2	3	4	3	2										

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HINT – SHEET

1. Ans (4)

$$PV = \frac{m}{M}RT$$

$$P \propto m.T$$

$$\frac{P}{P/2} = \frac{28}{m} \times \frac{330}{300}$$

$$m = \frac{28}{2} \times \frac{33}{30} = \frac{154}{10}$$

$$\text{Leaked mass} = 28 - \frac{154}{10} = \frac{126}{10} = \frac{63}{5} \text{ gm}$$

2. Ans (2)

$$dQ = dU + dW$$

$$dU = 0$$

$$dQ = dW = 600 \text{ J}$$

3. Ans (3)

From given curve,

Melting point for A = 60°C

and melting point for B = 20°C

Time taken by A for fusion = (6 – 2) = 4 minute

Time taken by B for fusion = (6.5 – 4) = 2.5 minute

$$\text{Then } \frac{H_A}{H_B} = \frac{6 \times 4 \times 60}{6 \times 2.5 \times 60} = \frac{8}{5}$$

4. Ans (4)

Forces acting on the joint are equal.

$$\therefore y = \frac{\text{Stress}}{\text{Strain}} = \frac{F}{A \times \text{Strain}}$$

$$\therefore F = Ay \times \text{Strain}$$

$$\text{where thermal strain} = \frac{\Delta \ell}{\ell} = \alpha \Delta T$$

$$\therefore F_1 = F_2$$

$$\Rightarrow A_1 y_1 \alpha_1 \Delta T = A_2 y_2 \alpha_2 \Delta T$$

$$\Rightarrow \frac{A_1}{A_2} = \frac{\alpha_2 y_2}{\alpha_1 y_1}$$

5. Ans (4)

The free body diagram of two parts are shown in figure



Total elongation is

$$\Delta l = \Delta l_1 + \Delta l_2 = \frac{3F \left(\frac{2L}{3} \right)}{AY} + \frac{2F \left(\frac{L}{3} \right)}{AY}$$

$$\Delta l = \frac{8FL}{3AY}$$

6. Ans (4)

$$\rho(\pi R^2 h)g = \sigma(\pi R^2 h_0)g + T(2\pi R)$$

$$h = \frac{\sigma}{\rho} h_0 + \frac{2T}{\rho R g} \Rightarrow h = 8 \text{ cm}$$

7. Ans (1)

$$\begin{aligned} n^2 &= n^1 \left[\frac{M_1}{M_2} \right]^a \left[\frac{L_1}{L_2} \right]^b \left[\frac{T_1}{T_2} \right]^c \\ &= 1 \times \left[\frac{10\text{g}}{1\text{kg}} \right]^1 \left[\frac{10\text{cm}}{1\text{m}} \right]^1 \left[\frac{0.1\text{s}}{1\text{s}} \right]^{-2} \\ &= \left[\frac{10\text{g}}{1000\text{g}} \right]^1 \left[\frac{10\text{cm}}{100\text{cm}} \right]^1 \left[\frac{1\text{s}}{10\text{s}} \right]^{-2} \\ &= \frac{1}{100} \times \frac{1}{10} \times 100 \\ &= \frac{1}{10} \text{ Newton} \end{aligned}$$

8. Ans (2)

$$\vec{A} = 10 \cos 53^\circ \hat{i} + 10 \sin 53^\circ \hat{j}$$

$$= 6\hat{i} + 8\hat{j}$$

$$\vec{B} = -10 \cos 37^\circ \hat{i} + 10 \sin 37^\circ \hat{j}$$

$$= -8\hat{i} + 6\hat{j}$$

$$\vec{C} = -10\hat{j}$$

$$\vec{R} = \vec{A} + \vec{B} + \vec{C} = -2\hat{i} + 4\hat{j}$$

9. Ans (1)

We consider the boat in the frame of reference of the air. Since we have the wind measurements, we can find the displacement of the boat in the frame of reference of the air. We'll be taking south and east as positive.

During the first segment we have the wind blowing east, so in the frame of reference of the air, the boat is displaced east by :

$$x_1 = 60v_1t_1$$

The 60 is there for a unit conversion of the time to seconds.

We do the same for the second leg, however we must account for the fact that the wind blows southeast, so there is both a southern displacement and an eastern displacement.

$$x_2 = 60v_2 \cos(\pi/4)t_2 = 30\sqrt{2}v_2t_2$$

$$y_2 = 60v_2 \sin(\pi/4)t_2 = 30\sqrt{2}v_2t_2$$

Similarly for the last leg, we have that

$$x_3 = 60v_3 \cos(3\pi/4)t_3 = -30\sqrt{2}v_3t_3$$

$$y_3 = 60v_3 \sin(3\pi/4)t_3 = 30\sqrt{2}v_3t_3$$

The total southern displacement is

$$y = 30\sqrt{2}v_2t_2 + 30\sqrt{2}v_3t_3 \approx 955\text{m}$$

The total eastern displacement is

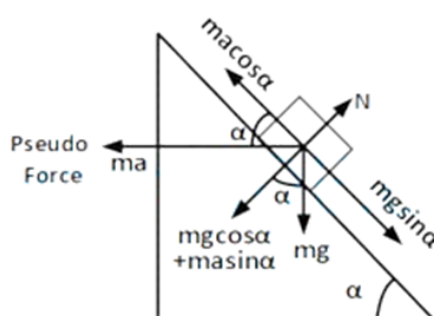
$$x = 60v_1t_1 + 30\sqrt{2}v_2t_2 - 30\sqrt{2}v_3t_3 \approx 3018\text{m}$$

Since the displacement in the lab frame is south 4000 m, we can find the displacement caused by the wind (in the lab frame) as 4000 - y south and -x east. From there we just divide by the total time to find the wind speed as it's constant, so

$$\begin{aligned} v_{\text{wind}} &= \frac{\sqrt{(4000 - y)^2 + (-x)^2}}{60(t_1 + t_2 + t_3)} \\ &= \frac{\sqrt{3045^2 + 3018^2}}{60} \approx 12 \text{ m/s} \end{aligned}$$

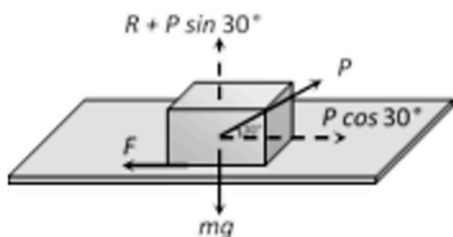
10. Ans (3)

Acceleration of the block is to the right. Pseudo force acting on the block to the left.



From diagram we can say, $ma \cos \alpha = mg \sin \alpha \Rightarrow a = g \tan \alpha$.

11. **Ans (1)**



$$R = mg - P \sin 30^\circ = mg - \frac{P}{2}$$

Friction between body and surface is given by,

$$F = \mu R = \mu \left(mg - \frac{P}{2} \right)$$

12. **Ans (3)**

Force on element towards axis

$$T - (T + dT) = -dT ; \quad -dT = (dm)\omega^2 x = \left(\frac{m}{l} dx \right) \omega^2 x$$

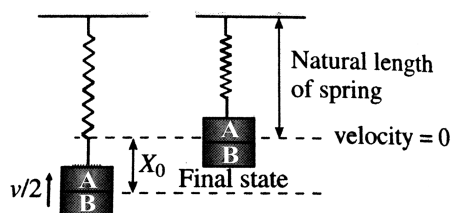
$$T = -\frac{m\omega^2}{l} \frac{x^2}{2} + \text{constant}; \text{ for } x=l, t=0 \text{ the constant} = \frac{1}{2} \frac{m\omega^2 l^2}{l}$$

$$T = \frac{1}{2} \frac{m\omega^2}{l} (l^2 - x^2)$$

13. **Ans (2)**

The initial extension in spring is $x_0 = \frac{mg}{k}$. Just after collision of B with A the speed of combined mass is $v/2$.

For the spring to just attain natural length the combined mass must rise up by $x_0 = \frac{mg}{k}$ (see the figure) and comes to rest.



Initial state

Applying conservation of energy between initial and final state:

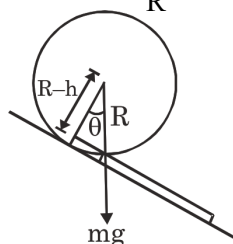
$$\frac{1}{2} 2m \left(\frac{v}{2} \right)^2 + \frac{1}{2} k \left(\frac{mg}{k} \right)^2 = 2mg \left(\frac{mg}{k} \right)$$

$$\text{Solving we get } v = \sqrt{\frac{6mg^2}{k}}$$

14. **Ans (4)**

The sphere is on the verge of toppling when a line of action of weight passes through the edge.

$$\cos \theta = \frac{(R-h)}{R} \Rightarrow h = R - R \cos \theta$$



15. **Ans (2)**

For D_2 , $V_P > V_n$

For D_1 , $V_P < V_n$

16. **Ans (4)**

Given circuit represent XOR.

17. **Ans (2)**

$$\left. \begin{aligned} 13 + 2e &= \lambda/2 \\ 41 + 2e &= \lambda \\ 69 + 2e &= 3\lambda/2 \end{aligned} \right\}$$

$$\lambda = 56 \text{ cm}$$

$$f = \frac{350}{56} \times 100 = 625 \text{ Hz}$$

18. **Ans (1)**

$$A_1 = 5 \quad A_2 = 3$$

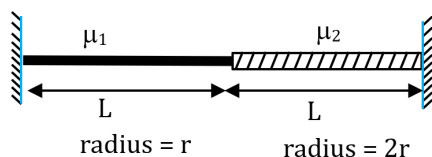
$$\Delta\theta = 2\pi(1.5) = 3\pi$$

$$A_{\text{net}} = \sqrt{A_1^2 + A_2^2 + 2A_1A_2 \cos(3\pi)}$$

$$= |A_1 - A_2|$$

$$= 2 \text{ cm}$$

19. **Ans (4)**



Let mass per unit length of wires are μ_1 and μ_2 respectively.

Materials are same, so density ρ is same.

$$\therefore \mu_1 = \frac{\rho \pi r^2 L}{L} = \mu$$

$$\text{and } \mu_2 = \frac{\rho 4\pi r^2 L}{L} = 4\mu$$

Tension in both are same = T , let speed of wave in wires are V_1 and V_2

$$V_1 = \sqrt{\frac{T}{\mu}} = V;$$

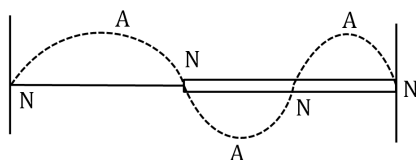
$$V_2 = \sqrt{\frac{T}{4\mu}} = \frac{V}{2}$$

So fundamental frequencies in both wires are

$$f_{01} = \frac{V_1}{2L} = \frac{V}{2L} \quad \text{and} \quad f_{02} = \frac{V_2}{2L} = \frac{V}{4L}$$

Frequency at which both resonate is L.C.M of both frequencies i.e. $\frac{V}{2L}$

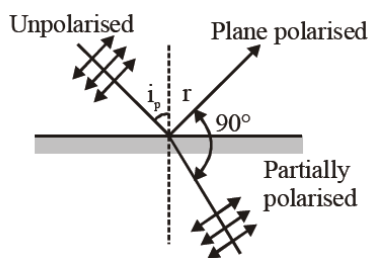
Hence no. of loops in wires are 1 and 2 respectively.



So, ratio of no. of antinodes is 1 : 2.

20. **Ans (3)**

From Brewster's law reflected ray is perpendicular to refracted ray.



The reflected ray is obtained is plane polarized having is electric vector in the plane of incidence.

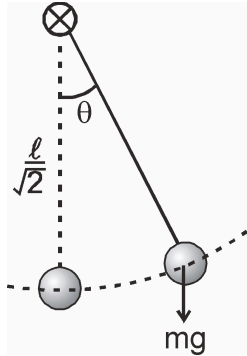
21. **Ans (4)**

$$\text{Resulting torque on the bob} = mg \frac{\ell}{\sqrt{2}} \sin \theta$$

$$\text{MI of bob about axis xy} = \frac{m\ell^2}{2}$$

For small angle θ

$$\alpha = \frac{\tau}{I} = \frac{\sqrt{2}g}{\ell} \theta$$



$$\omega = \sqrt{\frac{\sqrt{2}g}{\ell}} \Rightarrow T = 2\pi \sqrt{\frac{\ell}{\sqrt{2}g}}$$

22. **Ans (3)**

$$1 \text{ MSD} = x \text{ cm}$$

$$1 \text{ VSD} = \frac{(n-1)x}{n}$$

$$\begin{aligned} \text{LC} &= 1\text{MSD} - 1\text{VSD} = x - \frac{(n-1)x}{n} = \frac{x}{n} \\ &= x - \frac{n-x}{n} = \frac{x}{n} \text{ cm} \end{aligned}$$

23. **Ans (2)**

Radiation pressure

$$\begin{aligned} &= \frac{2P}{cA} = \frac{2 \times 3 \times 10^{-3}}{3 \times 10^8 \times 3 \times 10^{-6}} \\ &= 6.6 \times 10^{-6} \text{ N/m}^2 \end{aligned}$$

24. **Ans (2)**

$$\text{Energy } \frac{hc}{\lambda} = 4.5 \text{ eV}$$

Hence transition B

25. Ans (2)

Heavy water (D_2O) is used as a moderator in nuclear reaction.

26. Ans (2)

Theory

27. Ans (4)

$$E = G \left[\frac{M + \int_a^x \rho 4\pi (x^2 dx)}{x^2} \right]$$

$$= G \left[\frac{M + \frac{4\pi\alpha}{2} (x^2 - a^2)}{x^2} \right]$$

$$E = G2\pi\alpha + \frac{G}{x^2} [M - 2\pi\alpha a^2]$$

$$M = 2\pi\alpha a^2$$

$$\alpha = \frac{M}{2\pi a^2}$$

28. Ans (4)

$$v = k\pi\lambda$$



29. Ans (1)

$$R = 2\Omega, B = 0.020T, l = 32cm = 0.32m \text{ and } B = 8cm = 0.08m$$

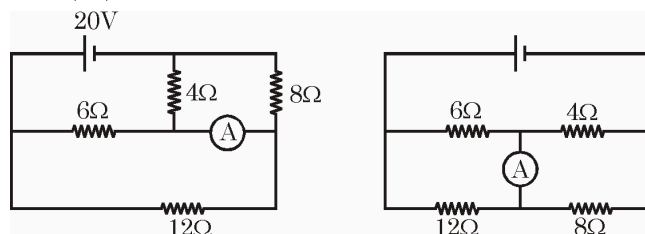
$$(a) F = ilB = 3.2 \times 10^{-5}N$$

$$= B^2 l^2 \frac{V}{R} = 3.2 \times 10^{-5}N$$

$$\Rightarrow \frac{(0.02)^2 \times (0.08)^2 \times v}{2} = 3.2 \times 10^{-5}N$$

$$\Rightarrow v = \frac{3.2 \times 10^{-5} \times 2}{6.4 \times 10^{-4} \times 4 \times 10^{-4}} = 25m/s$$

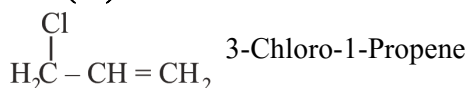
30. Ans (1)



$$\frac{6}{12} = \frac{4}{8}$$

So, it will act as wheatstone bridge, current through ammeter $\rightarrow$ zero

31. Ans (4)



32. **Ans (4)**

$$n \times \frac{5R}{2}(T_2 - 350) = -P_{\text{ext}} \left(\frac{nRT_2}{P_2} - \frac{nRT_1}{P_1} \right)$$

$$T_2 = 450 \text{ K}$$

$$W = nC_V(T_2 - T_1)$$

$$= 2 \times \frac{5R}{2} (450 - 350)$$

$$= 500 R$$

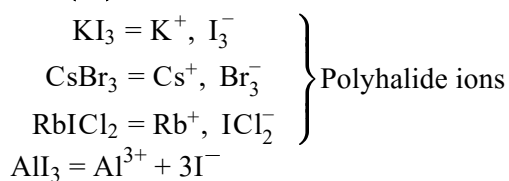
33. **Ans (2)**

3°-alcohol give turbidity immediate.

34. **Ans (4)**

CO shows synergic bonding.

35. **Ans (4)**



36. **Ans (1)**

$$k = \frac{2.303}{t} \log \frac{100}{100-x}$$

Substituting the values,

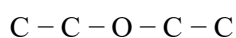
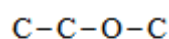
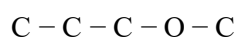
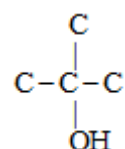
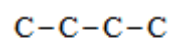
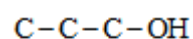
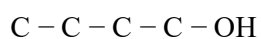
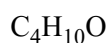
$$k = \frac{2.303}{32} \log \frac{100}{1}$$

$$k = 0.144 \text{ min}^{-1}$$

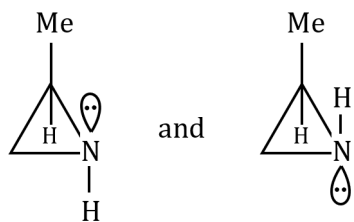
$$t = \frac{2.303}{0.144} \log \frac{100}{0.1}$$

$$t = 48 \text{ min}$$

37. **Ans (3)**



38. Ans (3)



Are Geometrical Isomers

39. Ans (1)

$$\frac{1}{\lambda} = R \times 2^2 \left(\frac{1}{1^2} - \frac{1}{2^2} \right) \Rightarrow \frac{1}{\lambda} = R \times 3$$

$$\frac{1}{\lambda'} = R \times 2^2 \left(\frac{1}{1^2} - \frac{1}{\infty^2} \right) \Rightarrow \frac{1}{\lambda'} = 4R$$

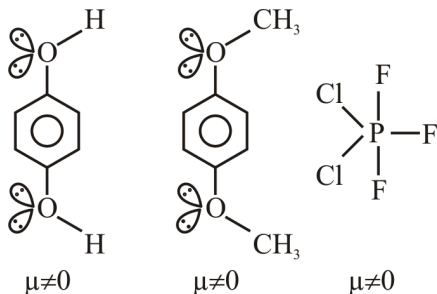
$$\lambda - \lambda' = \frac{1}{3R} - \frac{1}{4R} = \frac{1}{R} \left(\frac{1}{3} - \frac{1}{4} \right)$$

40. Ans (1)

Electron affinity of atom > Anion of that atom

$$X^+ > X > X^- > X^{2-}$$

41. Ans (4)



42. Ans (1)

$$M = \frac{\frac{20}{250}}{\frac{100}{1.25}} \times 1000 = \frac{80 \times 1.25}{100} = 1 \text{ M}$$

43. Ans (2)

$$n_{\text{MnO}_4^-} = \frac{0.4 \times 100}{1000} = 0.04$$

$$n_{\text{H}_2\text{O}_2} = \frac{5}{2} \times 0.04 = 0.1$$

$$M_{\text{H}_2\text{O}_2} = \frac{0.1}{50 \times 10^{-3}} = 2$$

$$\text{Volume st.} = 2 \times 11.35 = 22.7 \text{ V}$$

44. Ans (4)

On increasing temperature from 300 k to 330 k, rate becomes 8 times

$$\text{So } k_2 = 8k_1$$

$$\ln \left(\frac{k_2}{k_1} \right) = \frac{E_a}{R} \left(\frac{T_2 - T_1}{T_1 T_2} \right)$$

$$\ln(8) = \frac{E_a}{2} \left(\frac{330 - 300}{330 \times 300} \right)$$

$$\ln(23) = \frac{E_a}{2} \left(\frac{30}{330 \times 300} \right)$$

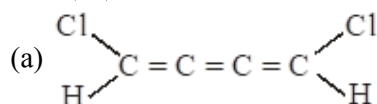
$$3 \ln(2) = \frac{E_a}{2} \left(\frac{1}{3300} \right)$$

$$E_a = 2 \times 3 \times 0.7 \times 3300 \text{ Cal/mol}$$

$$E_a = 13.86 \text{ kcal/mol}$$

45. Ans (3)

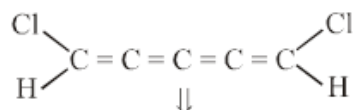
$$E^\circ_{\text{cell}} = 0.77 + 0.14 = 0.91 \text{ volt.}$$

46. Ans (1)


$\Rightarrow (=) \Rightarrow \text{odd.}$

(i) chiral atom is absent

(ii) always POS (it is molecular plane) is there because molecule is planar. It will not show O.I.

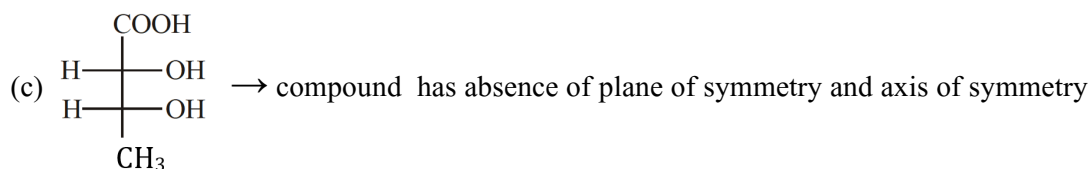


$(=) \Rightarrow \text{even}$

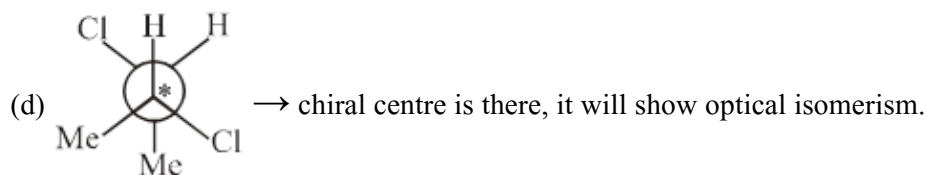
(b) POS, COSX

$\Downarrow$
Optically active compound

$\Downarrow$
It will show optical isomerism



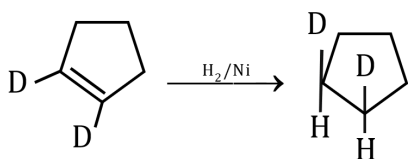
So, the given compound is optical active.



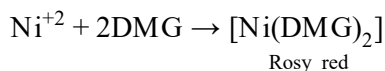
(A) is correct

47. **Ans (2)**

Catalytic hydrogenation proceeds through syn addition mechanism.



48. **Ans (4)**



49. **Ans (2)**

On reacting mixture ($\text{NH}_3 + \text{HCl}$)

Dalton's law cannot be applied.

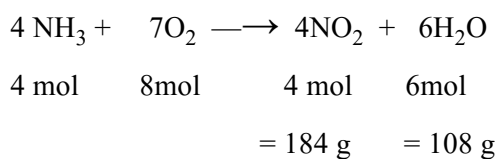
50. **Ans (3)**

Bead produced with CaSO_4 is colourless $\text{Ca}(\text{BO}_2)_2$, because Ca^{2+} is a colourless ion.

51. **Ans (2)**

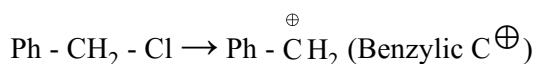
Reversing the second reaction and multiplying it by 2 and then subtracting the first reaction will give reaction 3.

52. **Ans (3)**

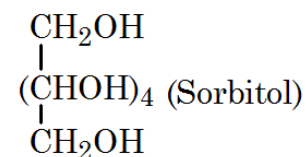
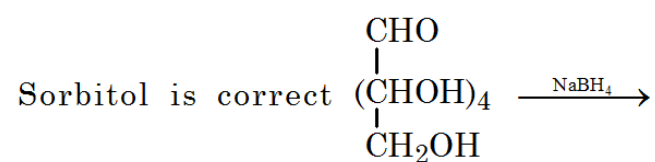


53. **Ans (3)**

Compound who form carbocation give ppt with AgNO_3



54. **Ans (1)**



55. **Ans (4)**

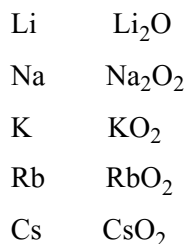
Equilibrium constant only depends on temp.

56. **Ans (2)**

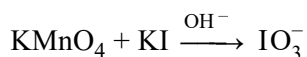
$$\begin{aligned}
 (K_{\text{SP}})_{\text{AgCl}} &= S^2 = 1.44 \times 10^{-4} \\
 S &= 1.2 \times 10^{-2} \text{M}
 \end{aligned}$$

57. **Ans (1)**

The stability of the oxide of alkali metals depends upon the comparability of size of cation and anion. Therefore the main oxide of alkali metals formed on excess of air are as follows :



58. **Ans (3)**



59. **Ans (1)**

(A)-(iii), (B)-(iv), (C)-(i), (D)-(ii)

60. **Ans (1)**

Novolac on heating with formaldehyde undergoes cross linking to form bakelite.

71. **Ans (3)**

Upon harvesting, cotton is gathered into bales; grain is gathered into shocks

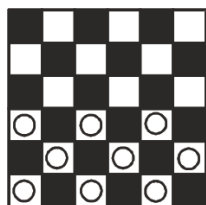
72. **Ans (3)**

Pattern is:

$$n : (n)^2 : n \times 10$$

So, here option (3) follows the same pattern.

73. **Ans (2)**



74. **Ans (1)**

By observation,



75. **Ans (4)**

90° Anti-clockwise movement of figure.

76. **Ans (4)**

$$5^3 - 5^2 = 125 - 25 = 100$$

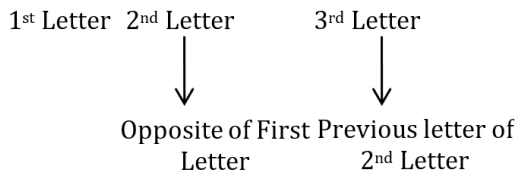
$$6^3 - 6^2 = 216 - 36 = 180$$

$$7^3 - 7^2 = 343 - 49 = 294$$

$$8^3 - 8^2 = 512 - 64 = 448 \neq 450$$

77. **Ans (2)**

All except option (2), follows this pattern.



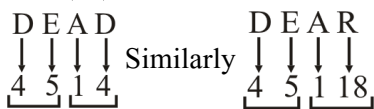
78. **Ans (3)**

In all other figures, none of the elements in at the central position and all the elements are arranged at the outer positions with the two types of elements appearing alternately.

79. **Ans (1)**

All the column's sum is 35

80. **Ans (3)**



81. **Ans (3)**

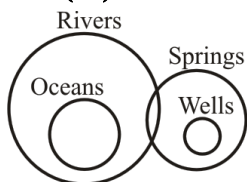
$3 \times 5 = 15$
 $15 + 10 = 25$
 $25 \times 4 = 100$
 $100 + 8 = 108 \neq 110$
 $108 \times 3 = 324$
 $324 + 6 = 330$
 $330 \times 2 = 660$
 $660 + 4 = 664$

82. **Ans (4)**

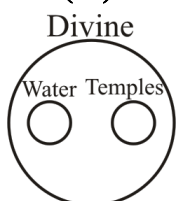
Three triangles and one triangle moving clockwise and anti-clockwise with flip of triangles.



83. **Ans (3)**

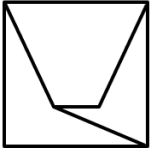


84. **Ans (4)**



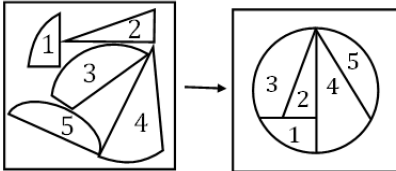
Neither conclusion I nor II follows.

85. **Ans (4)**



86. **Ans (2)**

By observation,



87. **Ans (1)**

In each column, three figures with one circle and one square each are present in such a way that three different patterns of both the circle and square are present in all columns.

88. **Ans (1)**

By observation,



89. **Ans (3)**

The passage deals with deteriorating living conditions in urban areas. Nothing about the birth rate is mentioned.

90. **Ans (3)**

According to the passage, a tiger takes to killing man only render it incapable to kill its natural prey - the animals in the forest.

91. **Ans (3)**

$$\begin{aligned} f'(x) &= (b-1)(b-2)(-2)\sin(2x) + (b-1) \\ &= (b-1)[1-2(b-2)\sin 2x] \neq 0 \\ b &\neq 1, \quad b \in \left(\frac{3}{2}, \frac{5}{2}\right) \end{aligned}$$

92. **Ans (4)**

$$y = x^2 - 5x + 5$$

$$\Rightarrow \frac{dy}{dx} = 2x - 5$$

slope of tangent at (2, 0)

$$m_1 = \left(\frac{dy}{dx}\right)_{(2,0)} = -1$$

slope of tangent at (3, 0)

$$m_2 = \left(\frac{dy}{dx}\right)_{(3,0)} = 1$$

$$\text{As } m_1 m_2 = -1$$

$$\Rightarrow \text{Angle between tangents is } \frac{\pi}{2}$$

93. **Ans (1)**

$$4 \times \left(\frac{\pi}{4} - \left(\int_0^1 (1-x^2) dx \right) \right)$$

94. **Ans (2)**

$$\begin{aligned} \log_{14} 84 &= \frac{\log_2 (4 \times 3 \times 7)}{\log_2 (2 \times 7)} \\ &= \frac{\log_2 4 + \log_2 3 + \log_2 7}{\log_2 2 + \log_2 7} = \frac{2 + \frac{1}{b} + a}{1 + a} = \frac{1 + 2b + ab}{b + ab} \end{aligned}$$

95. **Ans (2)**

$$\begin{aligned} (x+1)^{18} + (x+1)^{19} + (x+1)^{20} + \dots + (x+1)^{30} \\ = {}^{20}C_0 + {}^{21}C_1 + {}^{22}C_2 + \dots + {}^{30}C_{10} \\ = {}^{21}C_0 + {}^{21}C_1 + \dots + {}^{29}C_{10} + {}^{30}C_{10} = {}^{31}C_{20} \end{aligned}$$

96. **Ans (2)**

$$\begin{aligned} \text{(a) } f(0+h) &= \frac{\sin(\sin^2 h)}{(e^{\sinh} - 1)h} = 1 = f(0-h) \\ \text{(b) } f(a+h) &= \lim_{h \rightarrow 0} (1+h)(\operatorname{sgn}(0+h))^2 + h^2 = 1 \\ \text{LHL} &= -1 \end{aligned}$$

$$\text{(c) } \text{RHL} = \text{LHL} = \lim_{x \rightarrow 0} \frac{\ln(1 + \tan^2 x)}{\sin((e^x - 1)^2)} = 1$$

$$\text{(d) } \lim_{x \rightarrow 0} e^{\cos \operatorname{csc}(\sin^2 x)} = e^0 = 1$$

97. **Ans (2)**

$$\begin{aligned} z &= \left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6} \right)^5 + \left(\cos \frac{\pi}{6} - i \sin \frac{\pi}{6} \right)^5 \\ &= \cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6} + \cos \frac{5\pi}{6} - i \sin \frac{5\pi}{6} = 2 \cos \left(\frac{5\pi}{6} \right) \end{aligned}$$

98. **Ans (3)**

$$\begin{aligned} 2g_1 g_2 + 2f_1 f_2 &= c_1 + c_2 \\ 0 &= c - 4 \\ c &= 4 \end{aligned}$$

99. **Ans (2)**

$$\begin{aligned} \text{cross multiply} &\Rightarrow 3d(xy) + 2ydy - 2xdx = 0 \\ &\Rightarrow 3xy + y^2 - x^2 = c, \text{ passes through } (1,1) \\ &\Rightarrow c = 3 \\ &\Rightarrow x^2 - y^2 - 3xy + 3 = 0 \end{aligned}$$

100. Ans (1)

Let degree of $f(x) = n$

Degree of $f'(x) = n - 1$

degree of $f''(x) = n - 2$

$$\therefore f''(x) f'(x) = f(x)$$

$$\text{Then } (n - 2) + (n - 1) = n$$

$$2n - 3 = n$$

$$n = 3$$

$$\text{Now let } f(x) = ax^3 + bx^2 + cx + d$$

$$\text{Then } f'''(x) = 0$$

101. Ans (3)

$$I = \int_0^{\frac{\pi}{2}} \frac{10^{\sin x}}{10^{\sin x} + 10^{\cos x}} dx + \int_0^{\frac{\pi}{2}} \ln \tan x dx$$

$$\text{Let } I_1 = \int_0^{\frac{\pi}{2}} \frac{10^{\sin x}}{10^{\sin x} + 10^{\cos x}} dx$$

$$\text{and } I_2 = \int_0^{\frac{\pi}{2}} \ln \tan x dx$$

$$I_1 = \int_0^{\frac{\pi}{2}} \frac{10^{\sin x}}{10^{\sin x} + 10^{\cos x}} dx \quad \dots (1)$$

$$\Rightarrow I_1 = \int_0^{\frac{\pi}{2}} \frac{10^{\cos x}}{10^{\cos x} + 10^{\sin x}} dx \quad \dots (2)$$

Adding (1) & (2)

$$2I_1 = \int_0^{\frac{\pi}{2}} (1) dx$$

$$I_1 = \frac{\pi}{4} \text{ and}$$

$$I_2 = \int_0^{\frac{\pi}{2}} \ln \tan x dx = \int_0^{\frac{\pi}{2}} \ln \sin x dx - \int_0^{\frac{\pi}{2}} \ln \cos x dx = 0$$

$$\text{So } I_1 + I_2 = \frac{\pi}{4}$$

102. Ans (2)

$$\begin{aligned} \int_0^1 x(1-x)^9 dx &= \int_0^1 (1-x) x^9 dx \\ &= \int_0^1 (x^9 - x^{10}) dx = \left[\frac{x^{10}}{10} - \frac{x^{11}}{11} \right]_0^1 \\ &= \frac{1}{10} - \frac{1}{11} = \frac{1}{110} \end{aligned}$$

103. Ans (3)

The system of linear equation will have a non zero solution if $\begin{vmatrix} a^3 & (a+1)^3 & (a+2)^3 \\ a & (a+1) & (a+2) \\ 1 & 1 & 1 \end{vmatrix} = 0$

Now operate $c_2 - c_1$ and $c_3 - c_2$, then expand.

104. Ans (1)

Equation of chord with given middle point (x, y) is

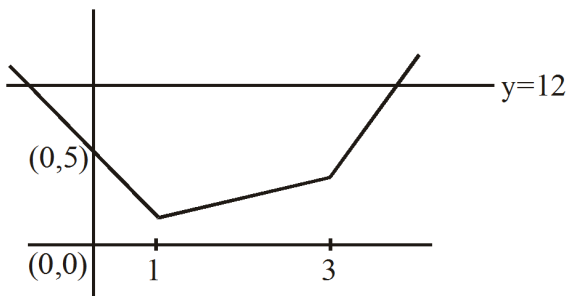
$$T = S_1 \Rightarrow \frac{xx_1}{4} + \frac{yy_1}{9} = \frac{x_1^2}{4} + \frac{y_1^2}{9} \quad \dots(1)$$

Compare (1) with $9x - 4y = 13$

$x_1 = 1$ and $y_1 = -1$ (i.e. $\alpha = 1$ & $\beta = -1$)

Hence, $(\alpha + \beta) = 0$

105. Ans (3)



$$|x - 3| + 2|x - 1| = 12$$

$\therefore$ given equation has two solutions.

106. Ans (3)

$$f(x) = [3 \sin^2 x + 3 \cos^2 x + \cos^2 x + 5]$$

$$f(x) = 3 + 5 + [\cos^2 x]$$

$$f(x) = 8 + [\cos^2 x] \Rightarrow f(x) \in \{8, 9\}$$

107. Ans (1)

$$\text{and } x^2 + y^2 = a^2$$

$$\Rightarrow x^4 - a^2 x^2 + c^4 = 0$$

this equation has 4 roots x_1, x_2, x_3 and x_4

$$x_1 + x_2 + x_3 + x_4 = 0$$

108. Ans (2)

$$\text{Put } (1 + \sin^2 x) = t \Rightarrow \sin 2x dx = dt$$

$$\text{Hence } \int \frac{\sin 2x}{1 + \sin^2 x} dx = \int \frac{1}{t} dt = \log(1 + \sin^2 x) + c.$$

109. **Ans (1)**

Put $\sin x = z$

$$\therefore \cos x \, dx = dz$$

$\therefore$ Given integral

$$\begin{aligned} &= \int z^5 \cdot \cos^2 x \, dz = \int z^5 (1 - z^2) dz \\ &= \int (z^5 - z^7) dz = \frac{z^6}{6} - \frac{z^8}{8} + c \\ &= \frac{\sin^6 x}{6} - \frac{\sin^8 x}{8} + c \end{aligned}$$

110. **Ans (3)**

$$\sum_{k=1}^n 2k = n(n+1)$$

$$\cot^{-1}(1 + n(n+1)) = \tan^{-1} \left(\frac{1}{1 + n(n+1)} \right)$$

$$\cot^{-1}(1 + n(n+1)) = \tan^{-1}(n+1) - \tan^{-1}(n)$$

$$\sum_{n=1}^{48} \cot^{-1}(1 + n(n+1))$$

$$= \sum_{n=1}^{48} \tan^{-1}(n+1) - \tan^{-1}(n)$$

$$= \tan^{-1}(49) - \tan^{-1}(1)$$

$$= \tan^{-1} \left(\frac{49-1}{1+49} \right)$$

$$= \tan^{-1} \left(\frac{24}{25} \right) = \cot^{-1} \left(\frac{25}{24} \right)$$

$$\cot \left(\sum_{n=1}^{48} \cot^{-1} \left(1 + \sum_{k=1}^n 2k \right) \right) = \cot \left(\tan^{-1} \frac{24}{25} \right) = \frac{25}{24}$$

111. **Ans (2)**

$$\lim_{x \rightarrow \infty} \sqrt{\frac{1 + \frac{\sin x}{x} + \frac{\sin^2 x}{x}}{1 + \frac{\cos x}{x} + \frac{\cos^2 x}{x}}} = \sqrt{\frac{1+0+0}{1+0+0}} = 1$$

112. **Ans (1)**

$$\lim_{x \rightarrow 0} \frac{(1+x)^{1/2} - (1+x)^{1/3}}{x}$$

use Binomial expansion

$$\begin{aligned} &\lim_{x \rightarrow 0} \frac{\left(1 + \frac{1}{2}x + \dots\right) - \left(1 + \frac{1}{3}x + \dots\right)}{x} \\ &= \frac{1}{2} - \frac{1}{3} = \frac{1}{6} \end{aligned}$$

113. **Ans (1)**

Let P : I become a teacher

q : I will open a school

Negation of $p \rightarrow q$ is $\sim(p \rightarrow q) = p \wedge \sim q$

114. **Ans (3)**

$$A B = \begin{bmatrix} 0 & c & -b \\ -c & 0 & a \\ b & -a & 0 \end{bmatrix} \begin{bmatrix} a^2 & ab & ac \\ ab & b^2 & bc \\ ac & bc & c^2 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = O$$

115. **Ans (4)**

$$A^{-1} = \frac{\text{adj}A}{|A|} = \frac{-1}{19} \begin{bmatrix} -2 & -3 \\ -5 & 2 \end{bmatrix} = \frac{1}{19} \begin{bmatrix} 2 & 3 \\ 5 & -2 \end{bmatrix} = \frac{1}{19} A$$

$$\Rightarrow 19 A^{-1} = A$$

116. **Ans (3)**

Let variance of first n positive integer be

$$\text{Var}(n) = \sigma^2$$

i.e. Variance of first n positive multiple of 5 will be

$$\text{Var}(5n) = 25\sigma^2$$

Variance of first n positive multiple of 10 will be $\text{Var}(10n) = 100\sigma^2$

$$\Rightarrow \frac{\text{Var}(5n)}{\text{Var}(10n)} = \frac{25\sigma^2}{100\sigma^2} = \frac{1}{4}$$

117. **Ans (2)**

$$\frac{1}{3} \left(\frac{1}{6} \right)^7 \cdot {}^4C_1 \cdot \left(\frac{1}{2} \right) \times \frac{1}{2} = \frac{2}{6^8}$$

118. **Ans (4)**

$$\text{Number of ways} = {}^5C_4 2^4 = 80$$

119. **Ans (4)**

$$y^2 = x - c$$

$\Rightarrow (0,0)$ lines on directrix

$$\Rightarrow \text{equation of directrix } x = c - \frac{1}{4}$$

$$\Rightarrow c = \frac{1}{4}$$

120. **Ans (1)**

A(2,1), B(6,-1) and C(2, -1)

The mid-point of AB is (4,0)

121. **Ans (1)**

$$y = 2x^2 - 4x + 10$$

$$= 2((x-1)^2 + 4)$$

$$y_{\min} = 8$$

122. **Ans (4)**

$$x = \frac{a}{2} \left(\frac{t+1}{t} \right) ; y = \frac{a}{2} \left(\frac{t-1}{t} \right)$$

$$\frac{2x}{a} - 1 = \frac{1}{t} ; \frac{2y}{a} - 1 = \frac{-1}{t}$$

eliminate 't'

123. Ans (2)

Let total number of families be x . Then according to question,

$$x = \frac{65x}{100} + 15000 - \frac{15x}{100}$$

$$\Rightarrow \frac{x}{2} = 15000 \Rightarrow x = 30,000$$

Thus, there are 30,000 families in the town.

124. Ans (4)

$$P = 1 + \frac{2}{3} + \frac{6}{3^2} + \frac{10}{3^3} + \frac{14}{3^4} + \dots + \infty$$

$$\frac{P}{3} = \frac{1}{3} + \frac{2}{3^2} + \frac{6}{3^3} + \frac{10}{3^4} + \dots + \infty$$

$$\frac{2P}{3} = 1 + \frac{1}{3} + \left[\frac{4}{3^2} + \frac{4}{3^3} + \frac{4}{3^4} + \dots + \infty \right]$$

$$\frac{2P}{3} = \frac{\frac{4}{3}}{1 - \frac{1}{3}}$$

$$\Rightarrow P = 3$$

125. Ans (4)

$$Q(2\lambda + 1, 4\lambda + 3, 3\lambda + 2) = 0 \text{ and } P(3, 8, 2)$$

$$\overrightarrow{PQ} = (2\lambda - 2)\hat{i} + (4\lambda - 5)\hat{j} + (3\lambda)\hat{k}$$

$$\overrightarrow{PQ} \cdot (3\hat{i} + 2\hat{j} - 2\hat{k}) = 0$$

$$3(2\lambda - 2) + 2(4\lambda - 5) - 2(3\lambda) = 0 \Rightarrow \lambda = 2$$

$$|\overrightarrow{PQ}| = \sqrt{2^2 + 3^2 + 6^2} = 7$$

126. Ans (2)

$$\vec{a} = \lambda \vec{b}$$

$$(x_1 - 2, 2) \equiv \lambda (1, y, z)$$

$$\frac{x}{1} = -\frac{2}{y} = \frac{2}{z} = \lambda$$

$$x = \lambda, y = -\frac{2}{\lambda}; z = \frac{2}{\lambda}$$

$$\hat{n} = \frac{\vec{c}}{|\vec{c}|} = \frac{\lambda\hat{i} - \frac{2}{\lambda}\hat{j} + \frac{2}{\lambda}\hat{k}}{\sqrt{\lambda^2 + \frac{4}{\lambda^2} + \frac{4}{\lambda^2}}}$$

$$\hat{n} = \frac{1}{\sqrt{\lambda^4 + 8}} (\lambda^2, -2, 2)$$

$$\hat{n} = \frac{1}{2\sqrt{2}} (0\hat{i} - 2\hat{j} + 2\hat{k})$$

$$\lambda = 2$$

$$\hat{n} = \frac{1}{\sqrt{24}} (4\hat{i} - 2\hat{j} + 2\hat{k}) = \frac{2\hat{i} - \hat{j} + \hat{k}}{\sqrt{6}}$$

and remaining options can not be obtained for any value of λ

127. Ans (3)

$$|\hat{a} + \hat{c}|^2 = |\sqrt{3}\hat{b}|^2$$

$$1 + 1 + 2 \cos \theta = 3$$

where θ = angle between $\hat{a}$ and $\hat{c}$

$$\Rightarrow \cos \theta = \frac{1}{2} \text{ So, } \theta = \frac{\pi}{3}$$

128. Ans (4)

$$\sin A \cos B \cos C + \sin B \cos C \cos A + \sin C \cos A \cos B$$

$$= \cos A \cos B \cos C (\tan A + \tan B + \tan C)$$

$$= \cos A \cos B \cos C \cdot \tan A \tan B \tan C$$

$$= \sin A \sin B \sin C$$

129. Ans (3)

$$\begin{aligned} & \frac{1}{1 + \sqrt{2}} + \frac{1}{\sqrt{2} + \sqrt{3}} + \frac{1}{\sqrt{3} + \sqrt{4}} + \dots \\ &= \frac{\sqrt{2} - 1}{\sqrt{2} - 1} + \frac{\sqrt{3} - \sqrt{2}}{\sqrt{3} - \sqrt{2}} + \dots + \frac{\sqrt{16} - \sqrt{15}}{\sqrt{16} - \sqrt{15}} \\ &= \sqrt{16} - 1 = 3. \end{aligned}$$

130. Ans (2)

Here $x_1 = 2, y_1 = 3$ and $\theta = 45^\circ$ Hence $\frac{x-2}{\cos 45^\circ} = \frac{y-3}{\sin 45^\circ} = r$

from first two parts $\Rightarrow x - 2 = y - 3 \Rightarrow x - y + 1 = 0$

Co-ordinate to point P on this line is $\left(2 + \frac{r}{\sqrt{2}}, 3 + \frac{r}{\sqrt{2}}\right)$.

If this point is on line $x + y + 1 = 0$ then

$$\left(2 + \frac{r}{\sqrt{2}}\right) + \left(3 + \frac{r}{\sqrt{2}}\right) + 1 = 0$$

$$\Rightarrow r = -3\sqrt{2} ; |r| = 3\sqrt{2}$$